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SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

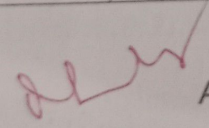
I K. Sandhiya / 192521318

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%	1	1	1	1
Debugging	25%	1	1	2	1
Optimization	20%	1	1	1	1
Analysis	20%	1	1	1	1
Code Quality	15%	1	1	1	1

25/30  Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

- Calculating the total number of usable host addresses after correction.
2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated

IPv4 Network Configuration

Given configuration:

- * IP Address: 192.168.10.65
- * Subnet Mask: 255.255.255.192 (126)
- * Default gateway: 192.168.10.1

Step 1: identify the subnet

A/26 subnet mask means:

* Block Size = 64

* Subnets are:

192.168.10.0 - 192.168.10.63

192.168.10.64 - 192.168.10.127

192.168.10.128 - 192.168.10.191

192.168.10.192 - 192.168.10.255

192.168.10.64 / 26

Step 2: Network Details

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Valid Host Range

first range = 192.168.10.65

Last host = 192.168.10.126

Step 3: Verify IP Address and gateway

IP address

192.168.10.65 in valid host range
Status Valid.

Default gateway

configured gateway = 192.168.10.1

The gateway = 192.168.10.1

~~192.168.100/26~~

The workstation belongs to:

192.168.10.64/26

Status X Invalid.

Step 4: configuration Errors:

* IP address belongs to 192.168.10.64/26

* gateway belongs to 192.168.10.0/26

* gateway must be in the same subnet
as the workstation.

Step 4: configuration

IP address belong to 192.168.10.64/26

Gateway belong to 192.168.10/26

Gateway must be in the same subnet as the workstation.

Step 5: Optimized configuration

use: IP Address: 192.168.10.66

Subnet Mask: 255.255.255.192

Default gateway: 192.168.10.65

(a)

IP = 192.168.10.70

gateway = 192.168.10.65

Both are valid.

Step 6: usable host addresses for /26

Number of host bit = 6

Total address

$$= 2^6 = 64$$

usable hosts

$$64 - 2$$

hosts.

2.

IPv6 configuration Debugging

Given:

Device A

2001:db8::ff00:8329/64

Device B

2001:db8:0:1::1/64

Step 1: Expand IPv6 Address

Device A

2001:0db8:0000:0000:000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network prefix

Prefix = /64

first 64 bits are network

Device A

Network Prefix

2001:db8:8:0:0::/64

Device B:

Network Prefix

2001:db8:0:0::/64

Step 4: Addressing Error

The IPv6 addresses are correctly configured.

Possible causes include:

firewall blocking ICMPv6

Neighbor Discovery failure

Duplicate address

Incorrect VLAN configuration

Step 5: Optimized configuration

Device A

2001:db8:0:0::10/64

Device B

2001:db8:8::0:20/64

Step 6: Available hosts

Host bits = 64

Total hosts

$= 2^{64}$

18,446,744,073,709,551,616

3. Subnet Allocation Optimization

Allocated

Step 5: optimized VLSM scheme

Largest Department = 126 (62 hosts)

Medium Department $\rightarrow$ 127 (30 hosts)

Small Department $\rightarrow$ 128 (14 hosts)

4. IPv6 configuration Debugging:

This question is identical to Question 2

Solution applies.

final answer:

Expanded Device A = 2001:0db8:0000:0000::f400:
0042:8329.

Expand Device B = 2001:0db8:0000:0000:0000:
0000:0001

Network Prefix = 2001:db8:0:0::/64

Error = NO Prefix mismatch; verify firewall

Neighbor Discovery, or VLAN configuration

Available Hosts = 2^{64} .

Routing Table Debugging

Destination Network

192.168.1.0/24

Next Hop

10.0.0.2

10.0.0.3

192.168.2.0/24

10.0.0.1

Default Route

Step 2: Select Next Hop

Next Hop

10.0.0.3

Step 3: Verify Destination Subnet

192.168.2.0/24

Host range

192.168.2.1 - 192.168.2.254

Destination

192.168.2.100

Step 5: Optimized Routing

Verify connectivity to 10.0.0.3

Ensure interfaces were active

Add redundant routers or dynamic routing (OSPF / RIP / EIGRP)

required.

Remove unused route to simplify forwarding

Step 6: Default Route

If no destination matches, the router forwards packets to:

10.0.0.1.

Default route = 10.0.0.1.